

Trip Itinerary

Generated by uber-polya

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Mode: Full Pipeline

1. Problem Formulation

Problem Type:	Find
Domain:	Integer Linear Programming
Named Problem:	Knapsack Problem

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	$\{0, 1\}$
n	number of binary variables = 10	$\mathbb{Z}^+$

Structure

Plan a one-day city trip. You have 10 attractions, each with an enjoyment score, visit duration (minutes), and entry cost (\$). You have 8 hours (480 minutes) and a \$100 budget. Select the subset of attractions that maximizes total enjoyment score without exceeding either budget. This is a variant of the 0/1 knapsack problem with two capacity constraints (time and money).

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

- (1) Multi-dimensional knapsack
- (2) Integer Linear Programming
- (3) Binary selection
- (4) Dual-budget trade-off
- (5) Sensitivity analysis

*0/1 knapsack generalized to two resource constraints
exact optimal solution via Branch & Bound
each attraction is either fully visited or skipped
balancing time and money simultaneously
how the itinerary changes if time or money changes*

Objective

$$\max \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach 0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: 0/1 Knapsack ILP

2. Solution

Answer:	Solution computed
Objective Value:	97.0
Optimal:	Yes
Feasible:	Yes
Algorithm:	0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)
Time:	0.0512s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Objective value: $z^* = 97.0$

Method: 0/1 Knapsack with two capacity constraints, formulated as Integer Linear Programming via PuLP/CBC.

- 10 binary variables (one per attraction: visit or skip)
- 2 capacity constraints (total duration ≤ 480 min, total cost $\leq \$100$)
- Objective: maximize sum of enjoyment scores for selected attractions

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Plan a one-day city trip.

The Answer

The optimal solution has objective value $z^* = 97.0$.

What This Means

- Optimal attraction selection maximizing total enjoyment under 8h and \$100
- Budget utilization bars for both time and money
- Per-attraction efficiency analysis (enjoyment per hour and per dollar)
- Sensitivity analysis across budget variations
- Independent verification of all constraints

0/1 Knapsack with two capacity constraints, formulated as Integer Linear Programming via PuLP/CBC.

- 10 binary variables (one per attraction: visit or skip)
- 2 capacity constraints (total duration ≤ 480 min, total cost $\leq \$100$)
- Objective: maximize sum of enjoyment scores for selected attractions

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution